

COSC 2671/3047 — Social Media and Network Analysis

Assignment 2 — Group Worksheet

Twitter/X Exodus to Mastodon: Analysing the Impact

Group 8 | May 2026

Team Members

Name	Student ID	Role
Bijo Babu Daniel	s4137699	Data Collection, Pre-Processing & Report Writing
Sachin Sathish Krishnan	s4140718	Network Analysis & Visualisation (Team Leader)
Serena Jibu Varghese	s4144470	NLP Analysis (Sentiment & Topic Modelling)

1. Team Project Plan

The project was divided into two weeks from 12 May to 24 May 2026, with the group presentation on 25 May. Work was split into four time blocks, with each member taking primary responsibility for their assigned component while contributing to shared tasks such as topic planning, report writing, and the final presentation.

Project Timeline

Period	Focus	Lead
12-14 May	Topic finalisation, API setup, pilot data collection, network environment setup	All
15-18 May	Full data collection, data cleaning, network construction, exploratory analysis	Bijo / Sachin
19-21 May	Centrality, community detection, Gephi, sentiment analysis, LDA preprocessing	Sachin / Serena
22-24 May	LDA topic modelling, network results, NLP results, discussion, final report	All
25 May	Presentation preparation and delivery	All

2. Weekly Timesheets

Bijo Babu Daniel (s4137699)

Date	Task	Hours
12 May	Topic finalisation and team planning	2
12 May	Mastodon API setup	2
13 May	Data collection script development	4
13 May	Pilot data collection run - 2 instances	2
14 May	Data quality check and CSV verification	2
14 May	Report - Introduction and Aim sections	2
15 May	Full 5-instance data collection (expanded hashtags)	3
16 May	Data cleaning - bots, duplicates, date filtering	3
17 May	Snapshot labelling and edge file construction	2
17 May	Report - Data Collection section	2
18 May	Report - Data Pre-Processing section	2
19 May	Report - Methodology section	3
20 May	Report review and cross-checking with results	2
21 May	Report - Exploratory Analysis section	2
22 May	Final report review and proofreading	2
23 May	Submission preparation and file organisation	2
24 May	Report - Discussion and Conclusion sections	2
25 May	Presentation preparation and delivery	2
	TOTAL	41

Sachin Sathish Krishnan (s4140718)

Date	Task	Hours
12 May	Topic finalisation and team planning	2
12 May	NetworkX environment setup and library installation	2
13 May	Review data collection output and verified edges	2
14 May	Network graph construction - build_graph function	2
15 May	User interaction network - full and per-snapshot	3
16 May	Instance federation network construction	2
17 May	PageRank and in-degree centrality computation	3
18 May	Betweenness centrality - NetworkX 2.5 compatibility fix	2
19 May	Louvain community detection and modularity scoring	4
20 May	Network cohesion metrics - WCC, SCC, reciprocity	3
21 May	Gephi visualisation - ForceAtlas2 layout and export	3
22 May	Network evolution table and snapshot comparison	2
23 May	Report - Network Analysis results section	3
24 May	Report - Discussion section contributions	2
25 May	Presentation preparation and delivery	2
	TOTAL	37

Serena Jibu Varghese (s4144470)

Date	Task	Hours
12 May	Topic finalisation and team planning	2
13 May	Review data collection output and verified toots	1
14 May	Initial insights - toot volume and snapshot charts	2
15 May	Join year distribution and exploratory visualisations	2
16 May	NLTK setup and preprocessing function development	2
17 May	VADER sentiment analysis across full dataset	3
18 May	Sentiment by snapshot and monthly arc visualisations	2
19 May	Sentiment by instance analysis	2
20 May	LDA preprocessing - tokenization, stopwords, lemmatize	3
21 May	LDA 5-topic model training and evaluation	3
22 May	Word cloud generation and topic evolution charts	3
23 May	Report - NLP Analysis results section	3
24 May	Report - Discussion section contributions	2
25 May	Presentation preparation and delivery	2
	TOTAL	32

3. Individual Reflection

Bijo Babu Daniel (s4137699)

This project was genuinely interesting, especially because the topic was something we stumbled upon rather than planned from the start. The idea of studying the Twitter exodus to Mastodon came up almost by accident, and it turned out to be a compelling research question. My main responsibility was data collection and preprocessing, which took longer than expected. There were many decisions to make around what to include and filter out - from API rate limiting to deduplication to choosing relevant hashtags. The approach was sometimes sluggish due to the variety of considerations involved. In hindsight, narrowing the scope earlier would have saved time, but the experience taught me a lot about real-world data collection challenges that theory alone cannot prepare you for.

Sachin Sathish Krishnan (s4140718)

Overall, I found this assignment complex but genuinely enjoyable, particularly the network analysis and visualisation components. Building the user interaction and instance federation networks with NetworkX was challenging, especially dealing with the NetworkX 2.5 betweenness centrality compatibility issue that required a workaround. Working through those problems was one of the most satisfying parts. The Gephi section was probably my favourite, seeing the hub-and-spoke structure emerge visually, with mho dominant at the centre and smaller clusters at the periphery, made the quantitative work feel concrete. The modularity score of 0.9175 was unexpected, and interpreting it alongside the high fragmentation rate was exactly the kind of nuanced thinking that made this project genuinely interesting.

Serena Jibu Varghese (s4144470)

My contribution was the NLP analysis - sentiment analysis using VADER and topic modelling using LDA. The analysis itself was interesting, particularly seeing the Fediverse Identity topic grow from 1.3% in Snapshot 1 to over 22% by Snapshot 3. However, the process was more time consuming than expected. Some techniques, particularly custom stopword development and coherence-based topic evaluation, went beyond what was directly covered in the course, requiring additional independent research. That said, it was a valuable learning experience and the final topics were clearly interpretable and relevant to the research question. If I were to do this again, I would allocate more time to preprocessing, as the stopword list had a significant impact on topic quality.